Open-Source Threat Intelligence & IOC Enrichment Dashboard

Project Report · Oct 4, 2026 · @Srivatsan Kumar

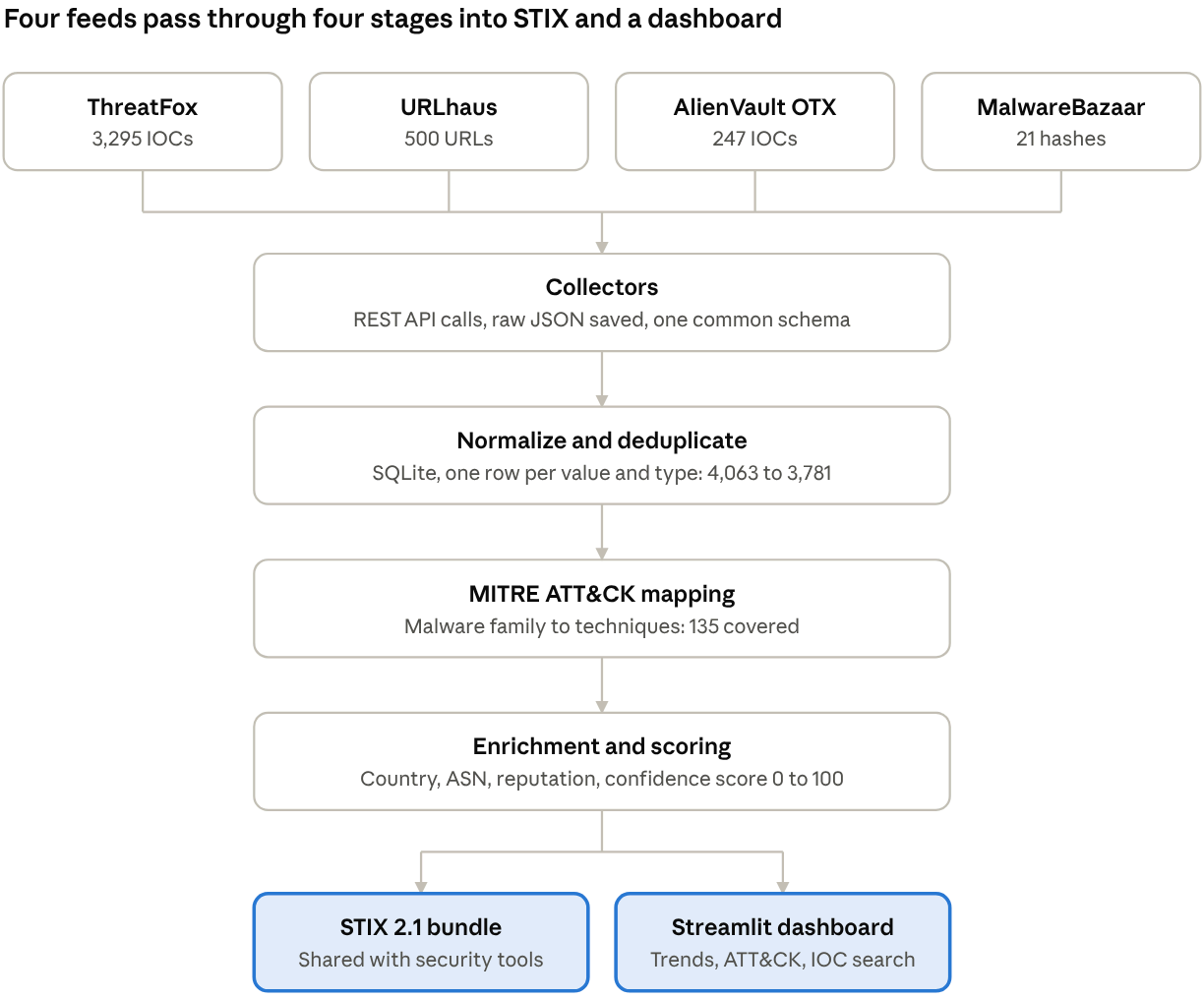
Overview

An automated Python pipeline that collects threat indicators (malicious IPs, domains, URLs and file hashes) from four free feeds, removes duplicates, enriches them, maps them to MITRE ATT&CK and shows them on a dashboard.

Feeds: ThreatFox, URLhaus and MalwareBazaar (abuse.ch), and AlienVault OTX.

System architecture

The pipeline runs as four sequential stages between the feeds and the two outputs, orchestrated by `pipeline.py`.



pipeline architecture · 4 feeds, 4 stages, 2 outputs

Each feed has its own collector; everything after collection works on one shared schema in SQLite.

Results

Metric	Value
Raw indicators collected	4,063
Unique indicators	3,781
Malware families	55
ATT&CK techniques	135

Metric	Value
High confidence (80+)	2,467

Most active threats: Mirai and Mozi (IoT botnets), AsyncRAT (remote access trojan), ClickFix (fake CAPTCHA scam) and AMOS (macOS stealer).

Challenges

- STIX export crashed on bad feed data; bad indicators are now skipped and logged.
- Free API rate limits; solved with batch lookups and a cap per run.

Future work

Expire old indicators, add a TAXII server, and move to PostgreSQL with Docker.

Technologies

Python, REST APIs, SQLite, MITRE ATT&CK, STIX 2.1, Streamlit, Plotly, Pandas. Source code is in the attached zip file.